A fixed completion on the compact Bob circle, and its exact defect

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This note replaces the residue-selected Bob parameter in the leading unit-L form by six fixed functions. An explicit source-only perturbation bound then gives a finite fixed testing family for the complete actual C1 parent on a complementary annulus. No certificate on the full testing domain is asserted: the last proposition identifies the negative term obstructing a three-event cap at the omitted corner. The source conventions are those of `pda4a21937f64`, `pb0d9d28d0849`, and `p23a094ff816d` in commit `fb58c232e374dadf3cbb5f23e7e7413d429de884`.

The six-function completion

Let

$$L > 0, \qquad 0 < r < 1, \qquad Lr < 1, \qquad C_0 = \frac{1}{2L(1-r)}, \quad A_0 = \frac{1+L(1-r)}{2Lr}, \quad a_0 = \frac{L}{2(1-Lr)}.$$

The inequalities on r and Lr are strict. All limits towards their openings are allowed; no value is assigned to a divergent coefficient at a literal zero denominator. Write

$$B = \frac{1+L(1-r)}{2Lr(1-r)}, \qquad \gamma = \frac{1+L(1-r)}{2L} = \frac{1}{2} + \frac{1}{4a_0}.$$

Let $c^2 + s^2 = 1$, with $c \geq 0$. The endpoints $c = 0$, $s = \pm 1$ are included. The four state variables P, X, Y, T below are independent. For finite physical angles the substitution is

$$c = (1+v^2)^{-1/2}, \quad s = vc, \quad P = uW, \quad T = cZ.$$

Identities on the independent raw variables therefore retain the full state polynomial; they impose neither a state graph nor a relation between the two state normals.

Lemma 1 (Residue-free Bob completion)**.** *For every source in the displayed domain,*

$$\begin{aligned} M(s) &= (C_0 + c^2/2)(P-Y)^2 + (A_0 + s^2/2)Y^2 + a_0(sY - T)^2 - sPT \\ &= a_0\left(T - sY - \frac{1-Lr}{L}sP\right)^2 + B(Y - rP)^2 + \gamma(cP)^2. \end{aligned} \tag{1}$$

Thus the fixed six-function family

$$(T, sY, sP, Y, P, cP)$$

has the positive source Gram matrix

$$\begin{pmatrix} a_0 & -a_0 & -1/2 & 0 & 0 & 0 \\ -a_0 & a_0 & 1/2 & 0 & 0 & 0 \\ -1/2 & 1/2 & 1/(4a_0) & 0 & 0 & 0 \\ 0 & 0 & 0 & B & -Br & 0 \\ 0 & 0 & 0 & -Br & Br^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \gamma \end{pmatrix}.$$

It has rank three, including at either Bob endpoint as an abstract feature Gram. Evaluation can lower the rank because the functions can become dependent there.

Proof. Put $k = C_0 + 1/2$. The source identities

$$C_0 + A_0 + 1/2 = B, \qquad k/B = r, \qquad kA_0/B = \gamma$$

reduce the claim to two successive completions of squares, first in T and then in Y. They also give the three rank-one summands of (1). No source or testing residue is chosen. □

A complete four-state annulus

The normalized complete signed core, in the same raw variables, is

$$\begin{aligned} K = {} & X^2 + c^2(P-Y)^2 + s^2Y^2 - 2sPT + 2ucX(sP - sY + T) \\ & + u^2s^2X^2 + u^2(sY - T)^2. \end{aligned} \tag{2}$$

Under $P = uW$, $T = cZ$, this is identically

$$\{c(P-Y) + usX - sZ\}^2 + (1+u^2)(sY - cZ)^2 + (X + uZ)^2 - (1+u^2)Z^2.$$

The whole leading selected quarter-parent is

$$S_0 = C_0(P-Y)^2 + A_0(1+u^2)Y^2 + a_0(1+u^2)(sY - T)^2 + K/2.$$

In particular, the independent X normal and all u-dependent terms remain present.

Proposition 2 (Explicit uniform leading-source reserve)**.** *Fix $0 < \ell \le U$, $0 < \tau \le 1$. If*

$$\ell \le L \le U, \quad |u| \le 1, \quad c^2 + s^2 = 1, \quad c \ge 0, \quad u^2 + c^2 \ge \tau^2,$$

the matrix of S_0 in (P, X, Y, T) satisfies

$$G \succeq \kappa I_4, \qquad \kappa = \frac{\min(\ell, 1)\tau^2}{4(7U^2 + 5U + 3)}.$$

This estimate is uniform through the three source openings $r \to 0$, $r \to 1$, and $Lr \to 1$ whenever they are compatible with the displayed source domain.

Proof. The restriction $T = 0$ is the strict three-variable sum of squares

$$\begin{aligned} & C_0(P-Y)^2 + A_0(1+u^2)Y^2 + a_0(1+u^2)s^2Y^2 \\ & \quad + \tfrac{1}{2}\{c(P-Y) + usX\}^2 + \tfrac{1}{2}X^2 + \tfrac{1}{2}s^2(1+u^2)Y^2. \end{aligned}$$

The exact determinant is

$$\det G = \frac{(1+u^2)^2\{Lc^2 + u^2s^2\}\{1 + L(1-r)\}^2}{16L^2r(1-r)(1-Lr)}.$$

It is positive unless $u = c = 0$. The strict upper state block and a one-field Schur complement therefore prove $G \succ 0$. This uses a complete state matrix, not a determinant without a normal-block check.

For the quantitative estimate put $t = u^2$, $q = c^2$, $e = 1 - r$ and $D = Lq + t(1 - q)$. Direct cofactors, with all opening denominators cancelled, give

$$\operatorname{tr} G^{-1} = \frac{2(L^2 N_2 + L N_1 + N_0)}{(1 + t) D (1 + Le)},$$

where

$$\begin{aligned} N_2 &= 1 + r^2 + eq\{1 + r(1 - q)\} + e(2 - qr)t + e^2(1 - q)t^2, \\ N_1 &= 2eq + \{2 - rq(3 - q)\}t + e(1 - q)t^2, \\ N_0 &= 1 + (2 - q)t. \end{aligned}$$

For $0 \le q, t \le 1$, $0 < r < 1$, these satisfy $0 \le N_2 \le 7$, $0 \le N_1 \le 5$, $0 \le N_0 \le 3$. Also

$$D \ge \min(\ell, 1)\{q + t(1 - q)\} \ge \tfrac{1}{2} \min(\ell, 1)(q + t).$$

Thus $\operatorname{tr} G^{-1} \le 1/\kappa$. The largest eigenvalue of G^{-1} is at most its trace, proving (2). □

Explicit fixed testing functions on that annulus

The same reserve admits actual source weights on an explicit domain. For positive C, A, a put

$$d_{\mathrm{src}} = |C/C_0 - 1| + |A/A_0 - 1| + |a/a_0 - 1|.$$

If the leading raw Gram is at least kI, the actual selected raw Gram is at least $\{k - (k + 3)d_{\mathrm{src}}\}I$. Indeed put $f = 1 - d_{\mathrm{src}}$, retain the nonnegative event excesses $C - fC_0$, $A - fA_0$, $a - fa_0$, and use the exact identity

$$S(C, A, a) = fS_0 + (1 - f)K/2 + \text{the three event excesses}.$$

The raw matrix of $K/2$ has absolute row sums at most $5/2$, hence operator norm at most 3. This proves the bound with both state normals retained.

Here is a finite construction, rather than an appeal to pointwise bounded-Gram positivity. Choose rational ℓ, U, τ in the displayed ranges, so that every fixed coefficient below is rational or algebraic. Set

$$k_* = \frac{\min(\ell, 1)\tau^2}{8(7U^2 + 5U + 3)}, \quad N = 4 + \frac{216}{k_*}, \quad h_* = \min(1, k_*/24), \quad B_* = N + 1.$$

The actual source admission is

$$C, A, a > 0, \qquad d_{\mathrm{src}} \le \frac{k_*}{2(k_* + 3)}.$$

It implies an actual raw reserve $k_*/2$ on the slightly larger annulus $u^2 + c^2 \ge \tau^2/2$. All infinitesimal unit-L sources with $\ell \le L \le U$ satisfy this admission, by the unchanged opening-safe ratios LA1/BE9. The finite statement here uses the displayed inequality itself and does not infer an explicit x, σ bound from an asymptotic assertion.

Clip the three source weights at N: $C' = \min(C, N)$, $A' = \min(A, N)$, $a' = \min(a, N)$. Retain the three nonnegative differences. Use the event coordinates

$$e = (P - Y, Y, sY - T, X)^T.$$

In these coordinates the clipped Gram has the exact form

$$G' = G_{\text{const}} + \sum_{j=1}^{7} \eta_j A_j, \qquad G_{\text{const}} = \operatorname{diag}(C', A', a', 1/2),$$

with testing monomials

$$\eta = (u^2, c^2, s^2, s, ucs, uc, u^2 s^2).$$

The matrices, with indices $1, \dots, 4$ and E_{ij} denoting matrix units, are

$$\begin{aligned}
A_1 &= \operatorname{diag}(0, A', a' + 1/2, 0), & A_2 &= E_{11}/2, \\
A_3 &= -(E_{12} + E_{21} + E_{22})/2, & A_4 &= (E_{13} + E_{31} + E_{23} + E_{32})/2, \\
A_5 &= (E_{14} + E_{41} + E_{24} + E_{42})/2, & A_6 &= -(E_{34} + E_{43})/2, \\
A_7 &= E_{44}/2.
\end{aligned}$$

All satisfy $\|A_j\| \le B_*$.

For clarity, the uniform clipping claim used here has a short proof. The core matrix in the event coordinates has row sums at most $5/2$, so its operator norm is at most 3. The inverse change of coordinates has squared Frobenius norm at most 6. At any point with $u^2 + c^2 \ge \tau^2/2$, the preceding proposition therefore gives the unclipped actual event-coordinate Gram at least $(k_*/12)I$. Split coordinates according as their source event weight exceeds N. The unchanged complementary principal block is at least $k_*/12$. The clipped large block is at least $N - 3$, and the off-diagonal block has norm at most 3. The inequality

$$6ab \le \frac{216}{k_*} a^2 + \frac{k_*}{24} b^2$$

proves $G' \succeq h_* I$. This argument uses only source clipping; the indices being clipped do not depend on a test.

Choose a fixed integer

$$n \ge \max\left(\frac{8}{\tau^2}, \frac{56 B_*}{h_*}\right), \qquad \delta = 4/n.$$

Take the finite algebraic grid

$$u_j = j/n, \qquad c_k = \cos\frac{k\pi}{2n}, \qquad s_k = \sin\frac{k\pi}{2n}, \qquad -n \le j, k \le n,$$

retaining centers $a = (u_j, c_k, s_k)$ with $u_j^2 + c_k^2 \ge \tau^2/2$. A chart consists of circle points within $1/n$ in each coordinate of its center. These charts cover the target annulus: nearest-grid errors are at most $1/(2n)$ in u and $\pi/(4n) < 1/n$ in either circle coordinate, and the change in $u^2 + c^2$ is at most $4/n$.

On a fixed chart put $b_j = \eta_j - \eta_j(a)$. Then $|b_j| \le \delta$. The exact matrix identity

$$\begin{aligned}
G' = {} & \big(G'(a) - 7\delta B_* I\big) \\
& + \frac{1}{2} \sum_{j=1}^{7} \left[(B_* I + A_j)(\delta + b_j) + (B_* I - A_j)(\delta - b_j)\right]
\end{aligned} \tag{3}$$

has only positive source matrices: its first block is at least $h_*/2$, and $B_* I \pm A_j \succeq 0$. Consequently the following *fixed*, source-independent testing functions suffice on that chart:

$$e_i, \quad e_i\sqrt{\delta + b_j}, \quad e_i\sqrt{\delta - b_j} \quad (1 \le i \le 4,\ 1 \le j \le 7),$$

together with

$$\sqrt{1+u^2}\,Y, \qquad \sqrt{1+u^2}\,(sY-T).$$

There are at most 62 functions per chart, and at most $(2n+1)^2$ charts. The excess $C-C'$ uses the already retained function e_1; the other two excesses use the last two functions. Their source Gram coefficients are $C-C'$, $A-A'$ and $a-a'$. The blocks in (3), the displayed A_j, and these excess coefficients are the complete evaluated matrix prescription. All centers are algebraic, all source entries are rational on the eight possible source clipping chambers, and every radical has a fixed constant δ. Factoring the positive matrices takes place only in the source field. These identities also hold over separately labelled real closed fields, chart by chart.

For the actual C1 source put $a=\alpha$. After the physical substitution $P=uW$, $T=cZ$, restore the three fixed testing functions

$$r_m = sW - cX + usY - ucZ, \quad r_D = c(uW-Y) + usX - sZ, \quad r_\beta = X + uZ.$$

Their source coefficients are respectively $m, D, \beta \geq 0$. The complete normalized parent, with $\lambda_{\mathrm{ev}} = n = 0$, is identically $S(C,A,\alpha) + mr_m^2 + Dr_D^2 + \beta r_\beta^2$. Thus at most 65 fixed functions per chart admit the *entire* actual C1 parent on this annulus. All $m, D, \beta \geq 0$ are allowed, including literal zero coefficients. All state variables and the literal Bob endpoints remain included. The restriction $|u| \leq 1$ and the omitted double corner remain part of the theorem's domain. Real and imaginary splitting handles complex states without new testing functions.

The exact obstruction for actual stationary weights

Proposition 3 (Below-wall defect and a full selected-parent witness)**.** *Let $C, A, a > 0$ be arbitrary and set*

$$k = C + 1/2, \quad B = C + A + 1/2, \quad g = \frac{kA}{B}, \quad \gamma = \frac{1}{2} + \frac{1}{4a},$$

$$\Delta = 8CAa - 4Ca - 1 - 2(C + A + a), \qquad d_* = \frac{\Delta}{8aB} = g - \gamma.$$

The moving Bob form obeys the exact identity

$$\begin{aligned}(C + c^2/2)(P-Y)^2 + (A + s^2/2)Y^2 + a(sY - T)^2 - sPT \\ = a\left(T - sY - \frac{sP}{2a}\right)^2 + B\left(Y - \frac{k}{B}P\right)^2 + gc^2P^2 + d_*s^2P^2. \end{aligned} \tag{4}$$

The leading weights have $\Delta = 0$. If $\Delta < 0$, the selected complete parent with weights (C, A, a) is negative at an explicit finite angle pair and a finite nonzero state.

Proof. Two square completions prove (4); the coefficient identity $8aB(g-\gamma) = \Delta$ is immediate by multiplication. For the last assertion retain the complete core (2) and put

$$X = 0, \quad P = 1, \quad Y = k/B, \quad T = sY + s/(2a), \quad E = A(k/B)^2 + F > 0, \qquad F = \frac{a + 1/2}{4a^2} > 0.$$

Its complete selected value is exactly

$$d_* + \gamma c^2 + Eu^2 - Fu^2c^2.$$

Choose

$$t = \frac{-d_*}{2(\gamma + E + 1)}, \quad u = c = \sqrt{t}, \quad s = \sqrt{1-t}.$$

Since $g > 0$, one has $0 < t < 1/2$. The selected value becomes

$$d_* \frac{\gamma + E + 2}{2(\gamma + E + 1)} - Ft^2 < 0.$$

The actual finite angle and state realization is $v = s/c$, $W = 1/u$, $X = 0$, $Y = k/B$, $Z = T/c$. Thus the witness uses the entire selected four-state polynomial, not a determinant or a state-normal truncation. □

For an actual finite noncommuting stationary contact of the full nonnegative-event parent, the existing strict three-event result (CAP14) gives $\Delta(C, A, \alpha) < 0$. Substituting $a = \alpha$ in (4) therefore exposes a strictly negative term which the leading source loses. The explicit witness disproves replacing the full source by that three-event cap. It is not a counterexample to the full parent: the omitted original events supply indispensable contact-scale contributions. The source identities BE3–BE9, the e-bound and SP6 have neither been altered nor replaced. Literal $D = 0$ or $\beta = 0$ alone do not remove the defect argument.

The annulus construction absorbs this defect only where the displayed strict reserve is available. The missing global step remains a fixed separated completion with the other original events at the double contact corner, projective Alice charts, and the remaining source scales and regions.